

About the Course

Distributed Systems

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Dipartimento di Informatica – Scienza e Ingegneria (DISI)
ALMA MATER STUDIORUM – Università di Bologna

Academic Year 2025/2026



- 1 Basics
- 2 Organisation
- 3 Materials
- 4 Exam



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Language I

We all live in a mixed-language environment

- e.g., I no longer have any courses in Italian
 - nor have I published a paper in Italian over the last fifteen years, at least
- the Department of Computer Science and Engineering (DISI) at UniBo started several international teaching initiatives in the last years
 - two of them in Cesena
 - obviously in English
- we do have *international students* here with us today
 - this year, we also have international students in Bologna from Bielefeld, Germany
 - and, *this course is in English because of that*

Language II

Our *lectures* here are given in *English*

- all of our *public conversations* will be held in English
 - in particular, both lectures and labs
 - and conversations via forums on Virtuale, too
- *private conversations*, instead, can be held in any language we possibly share
 - this includes the *final exam discussion*, too—even though that is public, in theory



Language III

The *material of the course* is in *English*

- all the *relevant literature* on computational systems is in *English*
 - the few translations concern just books, and are mostly stale
 - all the specifications, the tools, the languages, the manuals
 - most of the web material, including slides and videos from other courses
- so, these *slides are obviously in English*, too
- accordingly, any *student-produced material* should be in English, too
 - since it is supposed to be **shared** not just with professors, but with other students, too

Teachers

prof (M1) Andrea Omicini

- `mailto:andrea.omicini@unibo.it`
- `http://www.unibo.it/faculty/andrea.omicini/`
- `http://andreaomicini.apice.unibo.it/`

prof (M2) Giovanni Ciatto

- `mailto:giovanni.ciatto@unibo.it`
- `http://www.unibo.it/faculty/giovanni.ciatto/`
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tutor Matteo Magnini (Cesena)

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- `http://www.unibo.it/faculty/matteo.magnini/`
- `http://matteomagnini.apice.unibo.it/`

tutor ? (Bologna)



Pages I

Virtual Learning Environment (Virtuale)

- main link for the course

<https://virtuale.unibo.it/course/view.php?id=68868>

- all the information and the material for the course is there
- most of the interaction there, e.g.
 - the General Forum

<https://virtuale.unibo.it/mod/forum/view.php?id=2037608>

APICe Page of the Course

- our own page

<http://apice.unibo.it/xwiki/bin/view/Course/Ds2526>

- stable over the years

<http://apice.unibo.it/xwiki/bin/view/Courses/Series/Ds>

Pages II

- this course is (physically) is held at the Cesena Campus
- however, *this year* we also have students attending from Bologna, for the course “Distributed Software Systems”

APICe Page of the Proxy Bologna Course

- our own page

<http://apice.unibo.it/xwiki/bin/view/Course/Dss2526>

- students attending from Bologna can follow the course **online** via Virtuale
- we have destined some spaces in Bologna for students attending from there
- students attending from Bologna can contact the tutor for any issue
 - and the professors, of course

Virtuale Class

- the virtual room is available for Bologna students

https://teams.microsoft.com/l/meetup-join/19%3ameeting_ZTE2NDZjODUtM2JkYS00NWEyLWFkYjItMmQ1ZDUyMTI2NDJi%40thread.v2/0?context=%7b%22Tid%22%3a%22e99647dc-1b08-454a-bf8c-699181b389ab%22%2c%22id%22%3a%22c4e24c08-f59e-4091-a00d-05296e0c2f21%22%7d

- people with *permanent* or *recurrent issues* can contact us for a stable solution
- virtual access* for Cesena students can be granted in case of special needs
 - e.g., elections, illness, urgent family matters, ...
- the Cesena course is **not** an online course
 - we are using technologies to help us getting the best from the course
 - so, we expect Cesena students to *show up in class regularly*
 - and Bologna students to be *regularly attending online*, too

Recordings

- for all students possibly missing them for some good reasons, classes are *recorded* and made available via Virtuale
- not immediately, though
- as soon as possible, typically within a few days
- this is mostly designed for *students with permanent or recurrent issues*, preventing them from attending classes regularly
- recordings are **not** a substitute for attending classes



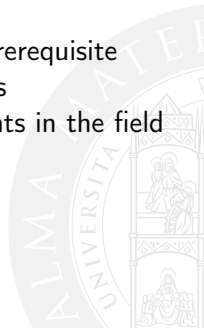
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Lessons

- classes are typically of three sorts
 - theory** classes ($M1$), aimed at introducing the fundamental issues of distributed systems
 - lab** classes ($M2$), where we experiment with technologies for distributed systems
 - preliminary** online classes, where we introduce some prerequisite knowledge and techniques required for labs
 - seminars** where we introduce some advanced contents in the field of distributed systems



Timetable I

General

- from 19 September 2025 to 19 December 2025
 - Friday, h. 9:00-11:00, Lab 3.1, Cesena Campus / Lab Ercolani, Bologna
 - Monday, h. 13:00-16:00, Room 2.4, Cesena Campus / Room Ercolani 3, Bologna

<https://www.unibo.it/en/teaching/course-unit-catalogue/course-unit/2025/526532/orariolezioni>



Timetable II

Module 1

- from 19 September 2025 to 15 December 2025
 - all Mondays
 - first Fridays

<https://www.unibo.it/en/teaching/course-unit-catalogue/course-unit/2025/526532/orariolezioni#537257>

Module 2

- from 17 October 2025 to 19 December 2025
 - most, last Fridays

<https://www.unibo.it/en/teaching/course-unit-catalogue/course-unit/2025/526532/orariolezioni#537258>

Timetable III

Any variation, exception extension, cancellation will be communicated via

- professors' AMS news page

<https://www.unibo.it/sitoweb/andrea.omicini/news>

<https://www.unibo.it/sitoweb/giovanni.ciatto/news>

- “News” section on VLE

<https://virtuale.unibo.it/mod/forum/view.php?id=2037607>



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Online

- typically, all the material for the course is made accessible through the Virtuale pages of the course
- and via the APICe pages, too
 - along with some additional resources
- typically available before the class
 - sometimes adapted / corrected just after



Slides

- the core material for the course is represented by the **slides**
- available as PDF via both Virtuale and APICe
- for theory, lab, and seminars
- ! slides represent the overall *reference* for the course



Books I

Disclaimer

- books are not mandatory
 - they are required neither to follow classes nor to pass the exam
 - yet, some of them are beautiful, and could be useful for a deeper understanding of many of the topics we are going to discuss



Books II

Some textbooks, anyway

- the basic textbook
 - original^[Tanenbaum and van Steen, 2017]
 - translated^[Tanenbaum and van Steen, 2007]
- other textbooks on
 - distributed systems design^[Coulouris et al., 2012]
 - distributed computing^[Kshemkalyani and Singhal, 2011]
 - distributed systems reliability^[Birman, 2005]
- general papers on
 - distributed computing models^[Lamport and Lynch, 1990]

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Exams

The final exam

- final exam consists of the oral discussion of a *project*
- after the *project artefacts check*
- once project artefacts are ok, students can sit the exam
- dates for exams are quite frequent
 - almost every week for those who complete the exam before next September
 - typically by appointment, via mail

Project Artefacts Check

- *projects* are checked on their *artefacts* (doc/code) before discussion
- once artefacts are ok, final discussion concludes all the matter
- professors and tutor only can decide when project artefacts are in a satisfactory state



What is a Project? I

Premises

- it should cover, or follow from, a specific topic of the course
- it may concern theoretical / technological / methodological aspects
- it may be developed by *individuals* or *groups*
- it has to be negotiated with the professor and/or the tutor, in order to ensure
 - its focus on distributed systems, as well as
 - its feasibility in a reasonable amount of time

What is a Project? II

Resources

- “Projects” section on Virtuale
 - with the Project Forum
<https://virtuale.unibo.it/mod/forum/view.php?id=2037611>
- “Projects” pages on APICe
 - this year
<http://apice.unibo.it/xwiki/bin/view/Course/Ds2526/Projects>
 - over the years
<http://apice.unibo.it/xwiki/bin/view/Courses/Series/Ds/Projects>

What is a Project? III

Action

- project activity can be started any time
 - the sooner the better, usually
- following and understanding classes and labs is practically required in the process
 - and typically **needs to show up** in the project artefacts
- professors and tutor will try and help you to understand where and how you should display your understanding of distributed systems topics in your project and – in particular – in your project artefacts
 - yet, the final responsibility comes down to you
 - and will be checked in the final discussion, in case some doubts arise

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References I

[Birman, 2005] Birman, K. P. (2005).

Reliable Distributed Systems. Technologies, Web Services, and Applications.

Springer New York

DOI:10.1007/0-387-27601-7

[Coulouris et al., 2012] Coulouris, G., Dollimore, J., Kindberg, T., and Blair, G. (2012).

Distributed Systems. Concepts and Design.

Pearson, 5th edition

<https://www.pearson.com/en-us/subject-catalog/p/distributed-systems-concepts-and-design/P200000003160/9780132143011>

[Kshemkalyani and Singhal, 2011] Kshemkalyani, A. D. and Singhal, M. (2011).

Distributed Computing: Principles, Algorithms, and Systems.

Cambridge University Press

<https://www.cambridge.org/us/universitypress/subjects/engineering/communications-and-signal-processing/distributed-computing-principles-algorithms-and-systems>

[Lamport and Lynch, 1990] Lamport, L. and Lynch, N. (1990).

Distributed computing: Models and methods.

In van Leeuwen, J., editor, *Handbook of Theoretical Computer Science, Volume B – Formal Models and Semantics*, chapter 18, pages 1157–1199. Elsevier

DOI:10.1016/B978-0-444-88074-1.50023-8

[Tanenbaum and van Steen, 2007] Tanenbaum, A. S. and van Steen, M. (2007).

Sistemi Distribuiti.

Pearson Education Italia, Torino, Italia, 2^a edition

http://hpe.pearsoned.it/scheda_opera.php?ID=2455



References II

[Tanenbaum and van Steen, 2017] Tanenbaum, A. S. and van Steen, M. (2017).
Distributed Systems. Principles and Paradigms.
Pearson Prentice Hall, 3rd edition
<https://www.distributed-systems.net/index.php/books/ds3/>

